

Java Program: StringIndexOutOfBoundsException Handling

1. Aim

To write a Java program that demonstrates a StringIndexOutOfBoundsException by accessing an invalid character position and handles the exception using a try-catch block.

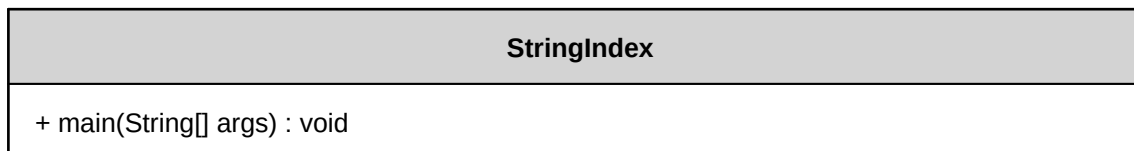
2. Description

A StringIndexOutOfBoundsException occurs when a program tries to access a character using an index that is outside the valid range of a string. In this program, the string **"Hello"** contains five characters, so its valid indexes are 0 to 4. The program tries to access index 10 using the **charAt()** method. This causes a StringIndexOutOfBoundsException, which is caught and handled by the catch block.

3. Algorithm

1. Start the program.
2. Enter the main method.
3. Create a String variable named text and assign the value "Hello".
4. Use the charAt() method to access the character at index 10.
5. Since index 10 is outside the valid range 0 to 4, a StringIndexOutOfBoundsException is generated.
6. Catch the exception using a catch block.
7. Display a message indicating that the string index is out of range.
8. Stop the program.

4. UML Diagram



5. Source Code with Comments

```
public class StringIndex {
    public static void main(String[] args) {
        try {
            // Create a string containing five characters
            String text = "Hello";

            // Try to access an invalid index
            // Valid indexes are 0 to 4
            char ch = text.charAt(10);

        } catch (StringIndexOutOfBoundsException e) {
            // Handle the invalid string index
            System.err.println(
                "Caught StringIndexOutOfBoundsException: "
                + "String index is out of range."
            );
        }
    }
}
```

6. Compilation and Execution Commands

```
javac StringIndex.java
java StringIndex
```

The first command compiles the Java source file. The second command executes the compiled program.

7. Output Screenshot

StringIndex - Console Output

```
Caught StringIndexOutOfBoundsException: String index is out of range.
```

8. Result

The Java program was successfully executed. It demonstrated a `StringIndexOutOfBoundsException` when an invalid index was used with the `charAt()` method, and the exception was successfully handled using a try-catch block.